

Module 2: POS Tagging & Hidden Markov Models

Penn Treebank Tagset, HMM 5-Tuple, Forward Algorithm, Viterbi Dynamic Trellis & CRFs

Module II: Part-of-Speech Tagging & Hidden Markov Models (HMM)

1. Linguistic Word Classes: Open Class vs. Closed Class Lexical Categories
2. Penn Treebank 45-Tag Tagset (NN, NNP, VB, VBD, JJ, RB, IN, DT)
3. Part-of-Speech (POS) Tagging Approaches (Rule-Based, Statistical, Neural)
4. Hidden Markov Models (HMM) 5-Tuple: States, Observations, Transitions, Emissions, Priors
5. The 3 Fundamental Computational Problems in HMMs (Evaluation, Decoding, Learning)
6. The Forward Algorithm for Sequence Likelihood ($O(N^2T)$ Dynamic Programming)
7. The Viterbi Algorithm for Optimal Hidden State Decoding (max Trellis Trace)
8. Complete Step-by-Step Viterbi Trellis Trace on Sentence "Janet will back the bill"
9. Baum-Welch (Expectation-Maximization) Unsupervised Learning Algorithm
10. Maximum Entropy Markov Models (MEMM) & The Label Bias Problem
11. Linear-Chain Conditional Random Fields (CRF) Globally Normalized Potentials
12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)

Topic 1 – 4: Part-of-Speech Tagging & HMM Mathematical Architecture

Part-of-Speech (POS) Tagging is the task of assigning a grammatical category label (e.g., Noun, Verb, Adjective, Preposition) to each word in an input text sequence:

$$\mathbf{w} = (w_1, w_2, \dots, w_T) \xrightarrow{\text{POS Tagging}} \mathbf{t} = (t_1, t_2, \dots, t_T)$$

Hidden Markov Model 5-Tuple $\lambda = (S, V, A, B, \pi)$:

1. $S = \{s_1, s_2, \dots, s_N\}$: Finite set of hidden states (e.g., POS tags).
2. $V = \{v_1, v_2, \dots, v_M\}$: Finite set of observable vocabulary symbols (words).
3. $A = \{a_{ij}\}$: State Transition Probability Matrix, where $a_{ij} = P(q_t = s_j \mid q_{t-1} = s_i)$.
4. $B = \{b_j(k)\}$: Emission Probability Matrix, where $b_j(k) = P(o_t = v_k \mid q_t = s_j)$.
5. $\pi = \{\pi_i\}$: Initial State Probability Distribution, where $\pi_i = P(q_1 = s_i)$.

Topic 7 & 8: The Viterbi Dynamic Programming Decoding Algorithm

Viterbi Recurrence Relations: Let $v_t(j)$ be the highest probability of any state sequence ending in state j at time t : 1.

Initialization ($t = 1$):

$$v_1(j) = \pi_j \cdot b_j(o_1), \quad \text{backpointer}_1(j) = 0 \quad (1 \leq j \leq N)$$

2. Recursion ($t = 2 \dots T$):

$$v_t(j) = \max_{i=1}^N \left(v_{t-1}(i) \cdot a_{ij} \right) \cdot b_j(o_t) \quad (1 \leq j \leq N)$$

$$\text{backpointer}_t(j) = \arg \max_{i=1}^N \left(v_{t-1}(i) \cdot a_{ij} \right)$$

3. Termination:

$$P^* = \max_{i=1}^N v_T(i), \quad q_T^* = \arg \max_{i=1}^N v_T(i)$$

4. Backtracking:

$$q_t^* = \text{backpointer}_{t+1}(q_{t+1}^*) \quad \text{for } t = T-1, T-2, \dots, 1$$

Topic 10 & 11: MEMMs vs. Linear-Chain CRFs (Label Bias Problem)

Model Class	Mathematical Formulation	Normalization & Key Theoretical Properties
1. Generative HMM	$P(\mathbf{w}, \mathbf{t}) = \prod_{i=1}^T P(t_i t_{i-1}) P(w_i t_i)$	Models joint probability; cannot easily integrate overlapping, non-independent features.
2. Discriminative MEMM	$P(\mathbf{t} \mathbf{w}) = \prod_{i=1}^T \frac{\exp(\mathbf{w}^T \mathbf{f}(t_i, t_{i-1}, \mathbf{w}, i))}{\sum_{t'} \exp(\mathbf{w}^T \mathbf{f}(t', t_{i-1}, \mathbf{w}, i))}$	Local Normalization at each step: Suffers from the <i>Label Bias Problem</i> where states with low-entropy transitions dominate regardless of input observations.
3. Linear-Chain CRF	$P(\mathbf{t} \mathbf{w}) = \frac{1}{Z(\mathbf{w})} \exp \left(\sum_{i=1}^T \sum_k w_k f_k(t_i, t_{i-1}, \mathbf{w}, i) \right)$	Global Normalization ($Z(\mathbf{w})$): Overcomes the Label Bias Problem completely by evaluating the total score of the entire sequence simultaneously.

Top BIT Mesra Exam Questions & Answers (Module II)

Q1. State the 3 fundamental problems of Hidden Markov Models and name the algorithms used to solve them. (6 Marks)

- 1. Evaluation Problem (Likelihood):** Given model λ and observation sequence O , compute $P(O | \lambda)$. Solved by the **Forward Algorithm** (or Backward Algorithm) in $O(N^2T)$ time.
- 2. Decoding Problem:** Given model λ and observation sequence O , find the optimal hidden state sequence Q^* that best explains O . Solved by the **Viterbi Algorithm** in $O(N^2T)$ time.
- 3. Learning Problem (Parameter Estimation):** Given observation sequence O , adjust model parameters $\lambda = (A, B, \pi)$ to maximize $P(O | \lambda)$. Solved by the **Baum-Welch (EM) Algorithm**.